

ITI Full Stack Project Requirements

Project

Smart Notes Workspace

Objective

Build a complete Full Stack application using React and Node.js. The application must include authentication, protected routes, CRUD operations, server communication, state management, and a responsive UI.

Frontend (React)

Use React, React Router, React Hook Form, Zod/Yup validation, TanStack Query, Redux Toolkit, Axios, Responsive Design, Dark/Light Theme.

Required Pages

Register, Login, Dashboard, Notes List, Create Note, Edit Note, Note Details, Profile.

Features

Register/Login using JWT.
Protected Routes.
Full CRUD for Notes.
Search.
Filter by Category/Status.
Sorting.
Pagination or Infinite Scroll.
Loading, Error and Empty States.
Confirmation before Delete.

Backend (Node.js)

Express.js, MongoDB, Mongoose, JWT Authentication, bcrypt, Validation, Environment Variables, Proper Error Handling.

Required APIs

POST /auth/register
POST /auth/login
GET /auth/me
GET /notes
GET /notes/:id
POST /notes
PATCH /notes/:id
DELETE /notes/:id

Database Model

User: name, email, password.

Note: title, content, category, tags, status, isPinned, userId, timestamps.

Bonus

Optimistic Updates, Debounced Search, Upload Profile Image, Markdown Support, Unit Tests, Swagger Documentation.

Submission

GitHub Frontend Repository.

GitHub Backend Repository.

README with setup instructions.

.env.example.

Postman Collection.

3-5 minute demo video.